↳ $X_1 X_2$ is typically correlated with X_1 & X_2, & so leaving them out tends to alter the meaning of interaction.

→ Non-Linear relationships ⇒ $mpg = \beta_0 + \beta_1 * \underbrace{horsepower}_{X_1} + \beta_2 * \underbrace{horsepower^2}_{X_2} + \varepsilon$.

But it is still a linear model.

⇒ Potential Prob's

① Non-linearity of data → Residual Plots can identify non-linearity.. (plot Residuals vs $\hat{y}_i$)

Check whether pattern in Residual plots or not.

② Correlat^n of Error terms. ε_i does not give info about ε_{i+1}.

If there is correlat^n among error terms then estimated SE will tend to underestimate the true SE. As a result, C.I & predict^n intervals will be narrower than they should be.

Ex. if we double the data (2n sample size) → estimated parameters would be same but C.I. would be narrower by a factor of $\sqrt{2}$.

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The SE, C.I. & hypothesis test rely upon this assumpt^n.

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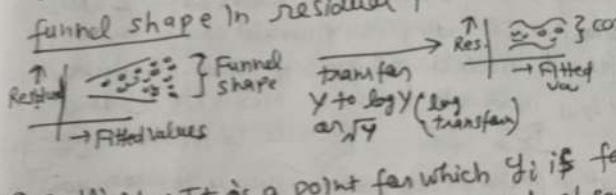

④ Outliers → It is a point for which y_i is far from predicted value $\hat{y}_i$.

↳ occurs from incorrect recording of observat^n.

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⑤ High Leverage Point → have an unusual value for x_i [not in normal range]

↳ leverage statistic $h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum (x_i - \bar{x})^2}$ [$\frac{1}{n} \le h_i \le 1$] & avg $(h_i) = \frac{p+1}{n}$,

& if $h_i \gg \frac{p+1}{n}$ then it is leverage point.

⑥ Collinearity = 2 or more variables are closely related.

Prob^m. it can difficult to separate out the individual effect on response & it reduces accuracy of $\beta_0, \beta_1 \ldots$ & SE grows. Identify using Correlat^n matrix

⇒ It is possible for collinearity to exist b/w 3 or more variables even if no 2 variables has high correlat^n = multicollinearity. Use VIF (Variance Inflat^n Factor)

~~VIF = Factor~~.

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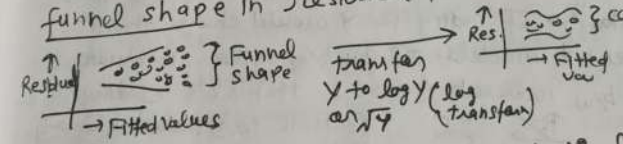

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~~VIF = Ratio~~.

⇒ VIF = ratio of variance of $\hat{\beta}_j$ when fitting the full model divided by variance of $\hat{\beta}_j$ if it fits on its own.. If VIF=1 means Complete absence of collinearity. VIF > 5 or VIF > 10 ⇒ Problematic

$VIF(\hat{\beta}_j) = \frac{1}{1-R^2_{X_j|X_{-j}}}$ → R^2 from a regression of X_j onto all of the other predictors.

→ In case of collinearity ⇒ Sol^n: ① drop one of the problematic variables

② Combine collinear variables together into a single predictor.

(*) In KNN Regression, we take mean of K too nearest neighbor points

(*) Value of K depends on bias-variance trade off

(*) KNN performance may decrease when dimension (no. of predictors) increases. because a given observat^n has no nearby neighbors in higher dimension = curse of dimensionality ⇒ very poor predict

(*) Parametric methods will tend to outperform non-parametric approaches when there is a small no. of observat^{n}s per predictor

(*) Classification

→ Response variable Y is qualitative (categorical). Ex. Eye Color..

→ Classifiers: Logistic Regression, LDA, QDA, Naive Bayes & KNN

→ Why not linear regression? → If coding for Y would change then it would produce different models. If gap b/w mild & moderate is similar to the gap b/w moderate & severe then 1, 2, 3 coding would be reasonable. (outside [0,1] interval in linear reg^n)

→ There are 2 reasons not to perform classification using a regression model!

(a) A regression method can't accommodate a qualitative response with >2 classes

(b) A " " will not provide meaningful estimates of $Pr(Y|X)$

→ We use logistic funct^n $p(X) = \frac{e^{\beta_0+\beta_1 X}}{1+e^{\beta_0+\beta_1 X}}$, For fitting, we use max^m likelihood

$p(X)/(1-p(X))$ = odds. (0 to ∞) & $\log(p(X)/(1-p(X))) = \beta_0 + \beta_1 X$ [log odds (logit) is linear in X]

→ We can also use non-linear least square to fit a model. but we use likelihood

→ Likelihood funct^n $\ell(\beta_0, \beta_1) = \prod_{i:y_i=1} p(x_i) \prod_{i':y_{i'}=0} (1-p(x_{i'}))$. The estimates $\hat{\beta}_0$ & $\hat{\beta}_1$ are chosen to maximize this likelihood funct^n.

→ The Z-statistic plays the same role as the t-statistic in linear reg^n o/p.

⇒ Multiple Logistic Reg^n ⇒ $\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$.

→ As in linear regression ... be quite different from ... when there is a correla... as cofounding.

→ In Multinomial Logis... ↳ we use softma...

→ $Pr(Y=k|X=x) =$ ... (Bayes' Theorem) Posterior Prob

↳ Estimating densit...

→ LDA

↳ Here we assume fo...

↳ Assump^n = share...

↳ discriminant func...

↳ For $p>1$, $X = (X_1, X_2$ ... class-specific mean vector...

↳ To indicate that a p... we write $X \sim N(\mu, \Sigma)$

$f(x) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp(-\frac{1}{2}$ ...

→ Bayes classifier ...

→ ROC curve is a grap... for all possible thresho...

↳ AUC = Are... left corner ... classifier th...

(*) Confusion Matrix

		p^*	N^* Predicted
Actual	P	TP	FN
	N	FP	TN

Precision $= \frac{TP}{TP+FP} = \frac{TP}{p^*}$

Recall $= \frac{TP}{p} = TPR$.

* LDA & Naive Bayes are sp... QDA " " " " not...

* Poisson Regression = when ...

$\log(\lambda(X_1, X_2, \dots X_p)) = ...$

→ As in linear regression, the result obtained using one predictor may be quite different from those obtained using multiple predictors, specially when there is a correlat^n among predictors. This phenomenon is known as cofounding.

→ In Multinomial Logistic Reg^n, treating K^{th} class as baseline is unimportant & we use softmax coding

→ $Pr(Y=k \mid X=x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)}$ [$f_k(x) = Pr(X \mid Y=k)$]

(Bayes' Theorem) Posterior Prob. Prior likelihood

↳ Estimating density funct^n $f_k(x)$ is much more challenging.

→ LDA

↳ Here we assume $f_k(x)$ is normal. $f_k(x) = \frac{1}{\sqrt{2\pi}\,\sigma_k} \exp\left(-\frac{1}{2\sigma_k^2}(x-\mu_k)^2\right)$ [μ_k & σ_k^2 are mean & variance of kth class]

↳ Assump^n = shared variance ⇒ $\sigma_1^2 = \dots = \sigma_K^2$.

↳ discriminant funct^n $\delta_k(x) = x\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$ is linear in x.

$p=1$

↳ For $p>1$, $X = (X_1, X_2, \dots)$ is drawn from a multi-variate Gaussian with class-specific mean vector & a common co-variance matrix. → Assume each predict^n follows normal

↳ To indicate that a p-dimensional r.v. X has a multivariate Gaussian dist^n we write $X \sim N(\mu, \Sigma)$, where $E(X) = \mu_p$ (vector) & $Cov(X) = \Sigma_{p\times p}$ matrix.

$f(x) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)$ & $\delta_k(x) = x^T\Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^T\Sigma^{-1}\mu_k + \log\pi_k$ is largest.

→ Bayes classifier uses a threshold 0.5.

→ ROC curve is a graphic tool for simultaneously displaying 2 types of errors for all possible thresholds. ROC (Receiver Operating Characteristic) comes from Comm^n Theory.

↳ AUC = Area under ROC curve. An ideal ROC curve hug the top left corner & has larger AUC. & better classifier. We expect a classifier that performs no better than chance to have AUC = 0.5.

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ROC: TPR (Sensitivity) (1-Type-II error) (Recall) (Power) → FPR (1-Specificity) Type-I error

$FPR = \frac{FP}{N}$

(*) QDA assumes each class has its own covariance matrix.

↳ High variance beca. $\frac{K\,p(p+1)}{2}$ parameters

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QDA " " " not " " " "

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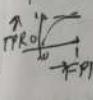

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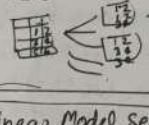
In Linear Regⁿ, $E(Y|X_1, \dots, X_p) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$.
" Logistic " , $E(Y|X_1, \dots, X_p) = (e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}) / (1 + e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p})$
" Poisson " , $E(Y|X_1, \dots, X_p) = e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}$.
link funct^n.
Belongs to Exponential dist^n Family. → others = Gamma dist^n, Negative Binomial

Resampling Methods.

↳ Computationally expensive because involves fitting same statistical method multiple times using diff. subsets of training data.
↳ Cross-Validat^n (CV) can be used to estimate the test error.
↳ Evaluating model's performance = Model Assessment & selecting proper level of flexibility = Model selection.
→ Bootstrap is used to provide a measure of accuracy of a parameter estimate.
⇒ CV Methods. ① Validation set Approach = Randomly dividing set of observat^n into 2 parts - training set & validat^n set (hold-out set). It overestimate test error.
② LOOCV (Leave One out CV) → Single observat^n = validat^n set & rest training set & repeat procedure n times → $CV = \frac{1}{n} \sum_{i=1}^{n} MSE_i$ → Less bias & overestimate test error.
③ k-Fold CV. = Randomly dividing observat^{n}s into k groups/folds. first fold is validation set & rest $k-1$ folds for fitting. $CV_{(k)} = \frac{1}{k} \sum_i MSE_i$
when $k = n \Rightarrow$ LOOCV. Usually $k = 5, 10$ (Computational advantage) • Bias-Var Tradeoff. (Less bias & variance)
⇒ Bootstrap. = is used to quantify uncertainty. It can also be used to estimate std. error of coeff. from Linear reg^n.

Ex.  $\rightarrow \hat{\alpha}^{*1}$, $\rightarrow \hat{\alpha}^{*2}$, ... $\Rightarrow SE_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1} \sum_{r=1}^{B} \left(\hat{\alpha}^{*r} - \frac{1}{B} \sum_{r'=1}^{B} \hat{\alpha}^{*r'}\right)^2}$
(with replacement sampling)

Linear Model Select^n & Regularizat^n.

$n >> p \Rightarrow$ Low Variance. & Otherwise there can be a lot of variability in the least squares fit, resulting in overfitting & there is no longer a unique least square coeff. estimates: ∞ many sol^{n}s.
⇒ Subset selection ① Best subset selection. : we fit a separate least square regress^n for each possible combinat^n of p predictors. $= \binom{p}{1} + \binom{p}{2} + \binom{p}{3} + \dots = 2^p$.
We use, C_p, AIC, BIC, adjusted R^2 to select best model among $M_0, M_1, \dots, M_p$
↳ Low RSS or high R^2 indicates low training error but we want low test error. So, we use C_p, AIC, BIC etc.
↳ In logistic reg^n, we use deviance instead of RSS. Smaller the deviance, better the fit.

② Stepwise Selec
statistical prob^n
↳ guided search
→ $C_p = \frac{1}{n}$ (RSS +
↓ unbiased estimate of test MSE.
→ BIC = $\frac{1}{n}$ (RSS
with many varia
→ Adjusted R^2 =

⇒ Shrinkage Metho
a technique that co
coeff. estimates to
reduces the variance
↳① Ridge Regression
by minimizing a sli
is a tunning para
of these 2 terms c
$\lambda \to \infty$ makes impact
ridge reg^n coeff.
generates only o
diff. set of coeff.
becaus it is the
↳ Ridge Reg^n coeff.
multiplying a give
So, standardize th
→ λ increases. →
↳② The Lasso (Linea
↳ Ridge reg^n inc
exactly zero. So, i
minimize RSS +
forcing some of th
So, it performs u

② Stepwise Selection – Best subset selection may also suffer from statistical probⁿ when p is large. → Forward → Backward → Mixed.
↳ guided search over model space. $p(p+1)/2$ models — more than — $1+p(p+1)/2$ models.

→ $C_p = \frac{1}{n}(RSS + 2d\hat{\sigma}^2)$ = AIC. & $\hat{\sigma}^2$ = unbiased estimate of σ^2 ($\sigma = Var(\epsilon)$)

C_p ↓ unbiased estimate of test MSE.

Penalty on training RSS. bcoz training error tends to underestimate test error. → Penalty increases as no. of predictors in model increases.

→ $BIC = \frac{1}{n}(RSS + \log(n)\, d\hat{\sigma}^2)$ → It places a heavier penalty on models with many variables & hence results in selectⁿ of smaller models than C_p.

→ Adjusted $R^2 = 1 - \frac{RSS/(n-p-1)}{TSS/(n-1)}$ [large adjusted R^2 is needed].

⇒ Shrinkage Methods We can fit a model containing all p predictors using a technique that constrains or regularizes the coeff. estimates or shrinks the coeff. estimates towards zero. Shrinking the coeff. estimates can significantly reduces the variance.

↳① Ridge Regression → Similar to least squares except that coeffs are estimated by minimizing a slightly diff. quantity. → Minimize $RSS + \underbrace{\lambda \sum_{j=1}^{p} \beta_j^2}_{\text{Shrinkage Penalty}}$ & $(\lambda \geq 0)$ → L2 Norm → No β_0

↳ Shrinking the estimates of β_j towards zero.

is a tunning parameter & controls the relative impact of these 2 terms on regression coeff. estimates.

$\lambda \to \infty$ makes impact of shrinkage penalty grows & the ridge regⁿ coeff. estimates will approach zero. Unlike least squares, which generates only one set of coeff. estimates, ridge regⁿ will produce a diff. set of coeff. estimates $\hat{\beta}_\lambda^R$, for each value of λ. We don't shrink β_0 because it is the mean value of response when $x_{i1} = x_{i2} = \dots = x_{ip} = 0$.

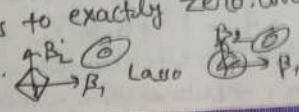
↳ Ridge Regⁿ coeff. estimates are not scale equivariant. i.e. it changes when multiplying a given predictor by a constant, it depends on λ too bcoz $\lambda X_j \hat{\beta}_j^R$. So, standardize the predictors to put it on the same scale.

→ λ increases → flexibility of fit decreases → Var ↓ bias ↑.

↳② The Lasso (Linear absolute shrinkage selectⁿ operator)

↳ Ridge regⁿ includes all p predictors in final model becaus coeff are not exactly zero. so, it creates the probⁿ of model interpretation.

minimize $RSS + \lambda \sum_{j=1}^{p} |\beta_j|$ [L1 Norm] → L1 Penalty has the effect of forcing some of the coeff. estimates to exactly zero when λ is large. So, it performs variable selectⁿ.



→ CV is used to find λ from a grid of values.

⇒ Dimensionality Redⁿ. methods. $Z_m = \sum_{j=1}^{p} \phi_{jm} X_j$. $[m < p]$

↳ 1st PC (principal component) directⁿ of the data is that along which the observat's vary the most. Projecting a point onto a line simply measures finding the locatⁿ on the line which is closest to points.

↳ 1st PC vector defines the line that is as close as possible to data.

↳ 2ndPC Z_2 is a linear combinatⁿ of variables that is concurrelated with Z_1 & has large variance subject to this constraint.

↳ Zero correlatⁿ condⁿ of Z_1 with Z_2 is equivalent to the condⁿ that the directⁿ must be perpendicular/orthogonal to 1st PC.

→ Principal Component Regressⁿ (PCR) uses $Z_1, Z_2 - Z_m$ for regressⁿ. m is chosen by CV.

→ PCA is feature extractⁿ method, not selectⁿ.

→ PCR & ridge regⁿ are closely related.

→ For PCR, we do standardizat o/w high variance variable will (PCA) play a larger role.

→ Partial Least Square (PLS) is a supervised alternat to PCR but not usefl.

→ When $p >> n$ then it is high-dimensional data & approaches such as least square linear regⁿ are not appropriate in this setting because of high variance (overfittⁿ).

→ Ridge Regⁿ, Lasso, PCR are useful in high dimensional settings.

→ In general, adding additional signal (noise) features that are truly associated with response will improve fitted model.

→ In High-dimensional setting, multicollinearity probⁿ is extreme.

⇒ Polynomial regressⁿ ⇒ add X, X^2, X^3 as predictor

→ Step functⁿ → Cut the range of a variable into K distinct regions & fit piecewise constant functⁿ.

→ Regression splines → Dividing range of X into K distinct regions & within each region, a polynomial functⁿ is fit to data. & they join by Knots at 25%, 50%, 75% quantiles smoothly (more knots when fun changes rapid)

→ Smoothing splines → minimize RSS with smoothness penality.

→ Basis Functⁿ. → $y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \dots + \epsilon_i$.

→ Knots = Points where coeff. changes. → Cubic splines with K knots has 4+K degree of freedom. → Natural spline = spline with boundary constraint

(*) Tree-Based Methods

↳ Combining a large n
in prediction accurac
↳ Decision Trees (DTs) c
↳ Pruning Our goal is
test error rate.. Cost
as a way to select smal
↳ α controls a trade-o
complexity & its fit to
↳ If we increase α f
→ Classificatⁿ error rat
training observat's in
→ Gini Index is a mea
→ Gini Index & Entro
→ Trees can be very
the data can cause
→ Trees can easily h
create dummy va
→ By aggregating m
random forests (RF)
→ Ensemble is an
blocks' models in
→ DT suffers from h
the variance of a m
→ Given a set of
variance σ^2, th
is given by $\frac{\sigma^2}{n}$
reduces varianc
from single train
→ The # of Trees B
large value of B
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of the observat
given bagged t

(*) Tree-Based Methods → simple & useful for interpretation.

↳ Combining a large no. of trees can often result in dynamic improvement in prediction accuracy at the expense of some loss in interpretat^n.

↳ Decision Trees (DTs) can be applied to both regression & classificat^n.

↳ Pruning Our goal is to select a subtree that leads to the lowest test error rate.. Cost complexity pruning (weakest link pruning) gives us a way to select small set of subtrees. $\underbrace{\sum_{m=1}^{|T|} \sum_{i:x_i \in R_m} (y_i - \hat{y}_{R_m})^2}_{\text{minimize.}} + \alpha \underbrace{|T|}_{\text{\# of Terminal nodes of Tree}}$

↳ α controls a trade-off b/w subtree's complexity & its fit to training data.

↳ If we increase α from zero, branches get pruned to find best subtree.

→ Classificat^n error rate is simply the fraction of the training observat^{n}s in that region that do not belong to most common class.

→ Gini Index is a measure of total variance across K classes.

→ Gini Index & Entropy are quite similar numerically.

→ Trees can be very non-robust. In other words, a small change in the data can cause a large change in final estimated tree.

→ Trees can easily handle qualitative predictors w/o the need to create dummy variables.

→ By aggregating many decision trees, using methods like bagging, random forests (RF) & boosting, predictive performance of trees can be improved.

→ Ensemble is an approach that combines many simple "building blocks" (weak learner) models in order to obtain a single & more powerful model.

→ DT suffers from high variance. Bootstrap aggregat^n (Bagging) reduces the variance of a ML method.

→ Given a set of n independent observat^{n}s $Z_1, Z_2, \ldots, Z_n$ each with variance σ^2, the variance of the mean $\bar{Z}$ of the observations is given by $\frac{\sigma^2}{n}$. In other words, averaging a set of observat^{n}s reduces variance. So, we can bootstrap, by taking repeated samples from single training data set & then train & predict & take average,

→ The # of Trees B is not a critical parameter with bagging; using a very large value of B will not lead to overfitting.

→ One can show that on average, each bagged tree makes use of $\frac{2}{3}$rd of the observations. Remaining $\frac{1}{3}$rd of observat^{n}s not used to fit a given bagged tree are refered as out of bag (OOB) observat^{n}s.